Stochastic Processes

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First, an example

Before any abstract definitions, picture a coin-flip walk. Let $X_1, X_2, \dots$ be independent with $X_i \in \{-1, +1\}$, each value having probability 1/2. Define the partial sums

$$S_n \;=\; X_1 + X_2 + \cdots + X_n, \qquad S_0 = 0.$$

Read aloud: "S sub n equals X sub one plus X sub two plus dot dot dot plus X sub n, with S sub zero equal to zero." Then $E[S_n] = 0$ and $\mathrm{Var}(S_n) = n$, so a typical sample path wanders to a distance of order $\sqrt{n}$ after n steps. Plotting many such paths, you see trajectories that jump up or down by exactly one at each tick and slowly diverge as $\sqrt{n}$. This is the simplest non-trivial stochastic process and the prototype for everything that follows.

1 Definitions

Definition 1 (Stochastic process). *Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(S, \mathcal{S})$ a measurable state space. A* stochastic process *indexed by a set T (typically $T = \mathbb{N}$ or $T = [0, \infty)$) is a collection of random variables*

$$X = \{X_t : t \in T\}, \qquad X_t : \Omega \to S \text{ measurable.}$$

Read aloud: *"X equals the collection of X sub t for t in T, where each X sub t is a measurable map from big Omega to S." For each fixed $\omega \in \Omega$, the function $t \mapsto X_t(\omega)$ is called a* sample path.

Definition 2 (Markov property). *A stochastic process $\{X_n\}_{n\in\mathbb{N}}$ has the* Markov property *(with respect to its natural filtration) if for every $n \in \mathbb{N}$ and every measurable set $A \subseteq S$,*

$$P\big(X_{n+1} \in A \;\big|\; X_0, X_1, \dots, X_n\big) \;=\; P\big(X_{n+1} \in A \;\big|\; X_n\big).$$

Read aloud: *"The probability that X sub n plus one lies in A given X sub zero, X sub one, through X sub n equals the probability that X sub n plus one lies in A given X sub n alone." Equivalently, given the present X_n, the future X_{n+1} is independent of the past $(X_0, \dots, X_{n-1})$.*

Definition 3 (Martingale). *Let $\{\mathcal{F}_n\}_{n\in\mathbb{N}}$ be a filtration on $(\Omega, \mathcal{F}, P)$. A process $\{X_n\}$ is a* martingale *with respect to $\{\mathcal{F}_n\}$ if*

1. *X_n is $\mathcal{F}_n$-measurable for every n (adaptedness);*
2. *$E[|X_n|] < \infty$ for every n (integrability);*
3. *$E[X_{n+1} \mid \mathcal{F}_n] = X_n$ almost surely (fair-game property).*

2 Simple Random Walk

Theorem 1. *The simple random walk* $S_n = X_1 + \cdots + X_n$ *with iid* $X_i \in \{-1, +1\}$ *each with probability* $1/2$ *satisfies* $E[S_n] = 0$ *and* $\mathrm{Var}(S_n) = n$.

Proof. Each summand X_i takes values ± 1 with equal probability, so

$$E[X_i] \;=\; (+1) \cdot \tfrac{1}{2} + (-1) \cdot \tfrac{1}{2} \;=\; 0,$$

Read aloud: "Expectation of X sub i equals plus one times one-half plus minus one times one-half, which equals zero." and

$$E[X_i^2] \;=\; (+1)^2 \cdot \tfrac{1}{2} + (-1)^2 \cdot \tfrac{1}{2} \;=\; 1.$$

Read aloud: "Expectation of X sub i squared equals plus one squared times one-half plus minus one squared times one-half, which equals one." Hence $\mathrm{Var}(X_i) = E[X_i^2] - (E[X_i])^2 = 1 - 0 = 1$.

By linearity of expectation,

$$E[S_n] \;=\; E\left[\sum_{i=1}^{n} X_i\right] \;=\; \sum_{i=1}^{n} E[X_i] \;=\; \sum_{i=1}^{n} 0 \;=\; 0.$$

Read aloud: "Expectation of S sub n equals expectation of the sum from i equals one to n of X sub i, which equals the sum from i equals one to n of expectation of X sub i, equal to zero."

Because the X_i are independent, all cross-covariances vanish: $\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$. The variance-of-a-sum identity therefore reduces to a sum of variances:

$$\mathrm{Var}(S_n) \;=\; \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) \;=\; \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j) \;=\; \sum_{i=1}^{n} 1 + 0 \;=\; n.$$

Read aloud: "Variance of S sub n equals variance of the sum of the X sub i, which equals the sum of variances plus twice the sum over i less than j of covariances, which collapses to the sum of ones, equal to n." This proves $E[S_n] = 0$ and $\mathrm{Var}(S_n) = n$. □

Remark. The standard deviation of S_n grows like $\sqrt{n}$, which is the diffusive scaling underlying Brownian motion (concept 32) and the noise schedule of diffusion-based generative models.